

2.5、三重积分换元法

设变换 $T : \begin{cases} x = x(u, v, w) \\ y = y(u, v, w) \\ z = z(u, v, w) \end{cases}$, 把 uvw 空间中的区域

Ω' 一对一的映射为 xyz 空间中的区域 Ω , 函数

$$x = x(u, v, w), \quad y = y(u, v, w), \quad z = z(u, v, w)$$

的一阶偏导数在 Ω' 内连续且函数行列式

$$J(u, v, w) = \frac{\partial(x, y, z)}{\partial(u, v, w)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix} \neq 0, (u, v, w) \in \Omega'$$

$$\text{则} \iiint_{\Omega} f(x, y, z) \mathrm{d}x \mathrm{d}y \mathrm{d}z$$

$$= \iiint_{\Omega'} f(x(u, v, w), y(u, v, w), z(u, v, w)) |J| \mathrm{d}u \mathrm{d}v \mathrm{d}w$$

柱坐标，球坐标是特殊的换元法.

对于柱坐标 $x = r \cos \theta, y = r \sin \theta, z = z$

$$J = \frac{\partial(x, y, z)}{\partial(r, \theta, z)} = \begin{vmatrix} \cos \theta & -r \sin \theta & 0 \\ \sin \theta & r \cos \theta & 0 \\ 0 & 0 & 1 \end{vmatrix} = r$$

对于球坐标 $x = r \sin \varphi \cos \theta, y = r \sin \varphi \sin \theta, z = r \cos \varphi$

$$J = \frac{\partial(x, y, z)}{\partial(r, \theta, \varphi)} = \begin{vmatrix} \sin \varphi \cos \theta & r \cos \varphi \cos \theta & -r \sin \varphi \sin \theta \\ \sin \varphi \sin \theta & r \cos \varphi \sin \theta & r \sin \varphi \cos \theta \\ \cos \varphi & -r \sin \varphi & 0 \end{vmatrix} = r^2 \sin \varphi$$

例1 求椭球面 $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ 所围成立体 Ω 的体积 V ($a, b, c > 0$).

解一 显然, 若将 Ω 变成单位球体, 计算将简单些, 为此考虑变换

$$u = \frac{x}{a}, v = \frac{y}{b}, w = \frac{z}{c}, \quad \text{即 } x = au, y = bv, z = cw,$$

$$\text{则有 } J = \frac{\partial(x, y, z)}{\partial(u, v, w)} = \begin{vmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{vmatrix} = abc$$

且 Ω 变为 $\Omega' = \{(u, v, w) | u^2 + v^2 + w^2 \leq 1\}$

$$\text{于是 } V = \iiint_{\Omega} dx dy dz = \iiint_{\Omega'} |J| du dv dw = abc \iiint_{\Omega'} du dv dw = \frac{4}{3} \pi abc.$$

解法二

作广义球坐标变换

$$T : \begin{cases} x = ar \sin \varphi \cos \theta \\ y = br \sin \varphi \sin \theta \\ z = cr \cos \varphi \end{cases}$$

则有 $J = abcr^2 \sin \varphi$

且 Ω 变为 $\Omega' = \{(\theta, \varphi, r) \mid 0 \leq \theta \leq 2\pi, 0 \leq \varphi \leq \pi, 0 \leq r \leq 1\}$

于是,

$$\begin{aligned} V &= \iiint_{\Omega} dV = \iiint_{\Omega'} abcr^2 \sin \varphi dr d\varphi d\theta \\ &= abc \int_0^{2\pi} d\theta \int_0^{\pi} \sin \varphi d\varphi \int_0^1 r^2 dr \\ &= \frac{4}{3} \pi abc. \end{aligned}$$

例2 求 $I = \iiint_{\Omega} z \, dx dy dz$, 其中 Ω 为由

$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \leq 1$ 与 $z \geq 0$ 所确定的区域.

解 作广义球坐标变换 $T : \begin{cases} x = ar \sin \varphi \cos \theta \\ y = br \sin \varphi \sin \theta \\ z = cr \cos \varphi \end{cases}$

于是, $J = abcr^2 \sin \varphi$

Ω 对应区域 $\Omega' = \left\{ (\theta, \varphi, r) \mid 0 \leq \theta \leq 2\pi, 0 \leq \varphi \leq \frac{\pi}{2}, 0 \leq r \leq 1 \right\}$,

$$I = \iiint_{\Omega} z \, dV = \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{2}} d\varphi \int_0^1 cr \cos \varphi \cdot abcr^2 \sin \varphi \, dr = \frac{\pi abc^2}{4}.$$

例3 计算 $I = \iiint_{\Omega} (x+y+z) \cos(x+y+z)^2 dx dy dz$

其中 $\Omega = \{(x, y, z) \mid 0 \leq x-y \leq 1, 0 \leq x-z \leq 1, 0 \leq x+y+z \leq 1\}$.

解 作变换
$$\begin{cases} u = x-y \\ v = x-z \\ z = x+y+z \end{cases} \quad \text{则有} \quad \frac{\partial(u, v, w)}{\partial(x, y, z)} = \begin{vmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 1 & 1 & 1 \end{vmatrix} = 3$$

且 Ω 变为 $\Omega' = \{(u, v, w) \mid 0 \leq u \leq 1, 0 \leq v \leq 1, 0 \leq w \leq 1\}$

$$\begin{aligned} I &= \iiint_{\Omega'} w \cos w^2 \left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| du dv dw \\ &= \frac{1}{3} \int_0^1 du \int_0^1 dv \int_0^1 w \cos w^2 dw = \frac{1}{6} \sin 1. \end{aligned}$$

例4 计算 $I = \iiint_{\Omega} (2x + 3y + 6z)^2 dx dy dz$

其中 $\Omega = \{(x, y, z) \mid 4x^2 + 9y^2 + 36z^2 \leq 36\}.$

解 由于 Ω 关于三个坐标平面对称, 故

$$\iiint_{\Omega} xy dx dy dz = \iiint_{\Omega} yz dx dy dz = \iiint_{\Omega} xz dx dy dz = 0$$

$$\begin{aligned} I &= \iiint_{\Omega} (4x^2 + 9y^2 + 36z^2) dx dy dz + 2 \iiint_{\Omega} (6xy + 12yz + 18xz) dx dy dz \\ &= \iiint_{\Omega} (4x^2 + 9y^2 + 36z^2) dx dy dz \end{aligned}$$

作广义球坐标变换 $T : \begin{cases} x = 3r \sin \varphi \cos \theta \\ y = 2r \sin \varphi \sin \theta \\ z = r \cos \varphi \end{cases}$

则有 $J = 6r^2 \sin \varphi$

且 Ω 变为 $\Omega' = \{(\theta, \varphi, r) \mid 0 \leq \theta \leq 2\pi, 0 \leq \varphi \leq \pi, 0 \leq r \leq 1\}$

于是,

$$\begin{aligned} I &= 36 \int_0^{2\pi} d\theta \int_0^\pi d\varphi \int_0^1 r^2 \cdot 6r^2 \sin \varphi dr \\ &= 216 \int_0^{2\pi} d\theta \int_0^\pi \sin \varphi d\varphi \int_0^1 r^4 dr \\ &= \frac{864}{5} \pi. \end{aligned}$$

例5 计算 $I = \iiint_{\Omega} \sqrt{x^2 + y^2} \, dx dy dz$

其中 $\Omega = \{(x, y, z) \mid x^2 + y^2 + (z - 1)^2 \leq 1\}.$

解 作坐标变换

$$T : \begin{cases} x = r \sin \varphi \cos \theta \\ y = r \sin \varphi \sin \theta \\ z = 1 + r \cos \varphi \end{cases}$$

则有 $J = r^2 \sin \varphi$

且 Ω **变为** $\Omega' = \{(\theta, \varphi, r) \mid 0 \leq \theta \leq 2\pi, 0 \leq \varphi \leq \pi, 0 \leq r \leq 1\}$

于是,

$$I = \int_0^{2\pi} d\theta \int_0^{\pi} d\varphi \int_0^1 r \sin \varphi \cdot r^2 \sin \varphi dr = 2\pi \int_0^{\pi} \sin^2 \varphi d\varphi \int_0^1 r^3 dr = \frac{\pi^2}{4}.$$

6 换成 2022, 换成 $2n$, 试一下!

例6 计算 $I = \iiint_{\Omega} x^6 dx dy dz,$

其中 $\Omega = \{(x, y, z) \mid x^2 + y^2 + z^2 \leq 1\}.$

解 作坐标变换 $T : \begin{cases} x = r \cos \varphi \\ y = r \sin \varphi \cos \theta \\ z = r \sin \varphi \sin \theta \end{cases}$

则有 $J = r^2 \sin \varphi$

且 Ω 变为 $\Omega' = \{(\theta, \varphi, r) \mid 0 \leq \theta \leq 2\pi, 0 \leq \varphi \leq \pi, 0 \leq r \leq 1\}$

于是, $I = \int_0^{2\pi} d\theta \int_0^{\pi} d\varphi \int_0^1 r^6 \cos^6 \varphi \cdot r^2 \sin \varphi dr$

$$= 2\pi \int_0^{\pi} \cos^6 \varphi \sin \varphi d\varphi \int_0^1 r^8 dr = \frac{4\pi}{63}.$$

练习 计算 $I = \iiint_{\Omega} x^6 dx dy dz ,$

其中 $\Omega = \left\{ (x, y, z) \left| \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \leq 1 \right. \right\}.$